Companion Derivation Note for the Lognormal Model Setting

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Purpose of this note

This document is written as a companion note to the simulation study plan, which aims to make the model-development logic explicit before jumping into the current proportional lognormal model setting.

The current simulation plan already presents a lognormal mixture model and a set of simulation scenarios. However, the rationale leundefinedading to this model needs to be shown more clearly. In particular, the plan should explain how the current lognormal working model was developed from the original framework of Vink et al. (2014), why the original normal/gamma assumptions may be insufficient for the current research question, and why the final model is set as a **structurally constrained lognormal mixture model.**

Each stage in this note addresses a limitation or question raised by the previous stage:

1. **Stage 1** reviews the original Vink framework and its normal/gamma path-mixture construction.
2. **Stage 2** clarifies the main limitation for the current project: PS is the primary estimand, while CP, PT, and PQ are nuisance path components that must be separated from PS.
3. **Stage 3** asks what happens if the latent primary-secondary serial interval is changed from normal/gamma to lognormal while keeping the original Vink transmission-path logic.
4. **Stage 4** explains why the exact lognormal extension becomes mathematically and computationally difficult.
5. **Stage 5** introduces a structurally constrained lognormal mixture model as a practical working model.
6. **Stage 6** discusses how the working-model parameters should be chosen after the basic distributional form has been selected. It first considers natural-scale moment constraints and then motivates the simpler log-scale parameterization.

The output of this note aims to provide a rationale for a mathematically reasonable model setting that should then be evaluated through the simulation study.

Notation and epidemiologic setting

Let X denote the serial interval (SI), defined as the time between symptom onset in a primary case and symptom onset in its secondary case.

For household data, the directly observed quantity is usually not the true infector–infectee SI. Instead, the observed quantity is the index case–to–case (ICC) interval, namely the time from symptom onset in the index case to symptom onset in another observed case in the same household.

Following Vink et al. (2014), an observed ICC interval may arise from several latent transmission paths:

- **CP**: co-primary path;
- **PS**: primary-secondary path;
- **PT**: primary-tertiary path;
- **PQ**: primary-quaternary path.

Let the mixture weights be

$$(w_{CP}, w_{PS}, w_{PT}, w_{PQ}), \qquad w_{CP} + w_{PS} + w_{PT} + w_{PQ} = 1.$$

The observed ICC density can be written generically as

$$f_Y(y) = w_{CP} f_{CP}(y) + w_{PS} f_{PS}(y) + w_{PT} f_{PT}(y) + w_{PQ} f_{PQ}(y).$$

The primary scientific target of the current project is the **PS serial interval distribution**. The other paths are nuisance transmission components that contaminate the observed ICC interval sample. The purpose of modeling CP, PT, and PQ is therefore to separate these nuisance components from the PS component as well as possible, so that the PS distribution can be estimated with less bias.

Throughout this note, θ and τ^2 are used only when describing the original Vink normal model. For the lognormal model, the notation is kept consistent with the main simulation study plan:

- Lognormal(μ, σ^2) means that $\log X \sim N(\mu, \sigma^2)$;
- μ is the log-scale location parameter for the PS component;
- σ is the common log-scale standard deviation;
- m and v are reserved for natural-scale means and variances.

Stage 1: The original Vink framework

1.1 Core idea of the Vink construction

The original Vink framework starts from one underlying SI distribution and derives the path-specific ICC distributions from it.

Under the baseline normal specification, write

$$X \sim N(\theta, \tau^2).$$

Let X_1, X_2, X_3 be independent and identically distributed copies of X. The four path-specific ICC intervals are then constructed as follows.

1.2 Co-primary path

For the co-primary path, two household members are assumed to have been infected independently by the same source outside the household. The observed ICC interval is

$$Y_{CP} = |X_1 - X_2|.$$

Under the normal assumption,

$$X_1 - X_2 \sim N(0, 2\tau^2),$$

so Y_{CP} has a half-normal type distribution.

1.3 Primary-secondary path

For direct primary-secondary transmission,

$$Y_{PS} = |X|.$$

Under the normal SI assumption, this gives a folded normal distribution with parameters θ and τ^2.

1.4 Primary-tertiary path

For a two-step transmission chain,

$$Y_{PT} = |X_1 + X_2|.$$

Under the normal assumption,

$$X_1 + X_2 \sim N(2\theta, 2\tau^2),$$

so Y_{PT} is again folded normal after taking the absolute value.

1.5 Primary-quaternary path

For a three-step transmission chain,

$$Y_{PQ} = |X_1 + X_2 + X_3|.$$

Under the normal assumption,

$$X_1 + X_2 + X_3 \sim N(3\theta, 3\tau^2),$$

so the corresponding ICC distribution is again folded normal after taking the absolute value.

1.6 Summary of the original normal model

The normal model is mathematically attractive because sums and differences of independent normal variables remain normal. This gives convenient closed-form path distributions for CP, PS, PT, and PQ.

A gamma alternative is also considered in the original Vink analysis when positivity and right-skewness are important. However, the same general issue remains for the current project: the classification of an observed ICC interval into CP, PS, PT, or PQ depends strongly on the path-specific distributions assumed in the mixture model.

Stage 2: why the current project needs to go beyond the Vink model

The current project is not only interested in reproducing the original Vink analysis. The main target is the PS serial interval distribution. CP, PT, and PQ are nuisance path components: they are not the primary scientific estimands, but they contaminate the ICC interval sample and must be separated from PS to obtain an accurate PS estimate.

In a mixture model, this separation is driven by the path-specific distributions specified for CP, PS, PT, and PQ. If these component distributions are poorly specified, the posterior path responsibilities can be distorted. As a result, ICC intervals that should mainly inform PS may be assigned to nuisance paths, or nuisance-path intervals may be misclassified as PS. Either situation can lead to biased estimates of the PS serial interval.

Therefore, the distributional assumption for the underlying PS serial interval is not only a technical detail. It directly affects path classification and PS recovery. From an epidemiologic perspective, the PS serial interval may be positive and right-skewed, because transmission timing is shaped by incubation period, infectiousness profile, symptom development, reporting delay, and behavioural response. A lognormal distribution is therefore a plausible alternative to the normal specification.

The next question is then:

What happens if the latent PS serial interval in the Vink construction is lognormal rather than normal?

The following stages explore this question step by step. The aim is not to jump directly to the final working model, but to show why such a working model becomes a reasonable approximation after considering the exact lognormal extension.

Stage 3: exact lognormal extension of the Vink construction

3.1 Replace the latent SI distribution

A direct extension is to keep the Vink transmission-path construction but replace the latent SI distribution by a lognormal distribution:

$$X \sim \mathrm{Lognormal}(\mu, \sigma^2),$$

meaning that

$$\log X \sim N(\mu, \sigma^2).$$

The lognormal density is

$$f_X(x; \mu, \sigma) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log x - \mu)^2}{2\sigma^2}\right\}, \qquad x > 0.$$

This retains the original Vink logic:

- the transmission-path distributions are induced from one common latent SI distribution;
- the observed ICC distribution is a mixture of the induced path-specific distributions;
- the higher-generation paths are generated through independent and identically distributed copies of the latent SI.

Let

$$X_1, X_2, X_3 \overset{i.i.d.}{\sim} \mathrm{Lognormal}(\mu, \sigma^2).$$

Under the exact lognormal extension, the four path-specific ICC variables are

$$Y_{CP} = |X_1 - X_2|, \qquad Y_{PS} = X_1, \qquad Y_{PT} = X_1 + X_2, \qquad Y_{PQ} = X_1 + X_2 + X_3.$$

Because a lognormal random variable is positive, the PS, PT, and PQ components do not require folding.

3.2 Path density for PS

For the PS path,

$$Y_{PS} = X_1.$$

Therefore, for $y > 0$,

$$f_{PS}(y) = f_X(y; \mu, \sigma) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log y - \mu)^2}{2\sigma^2}\right\}.$$

3.3 Path density for PT

For the PT path,

$$Y_{PT} = X_1 + X_2.$$

The density is the convolution of two lognormal densities:

$$f_{PT}(y) = \int_0^y f_X(x;\mu,\sigma) f_X(y-x;\mu,\sigma)\,dx, \qquad y > 0.$$

Substituting the lognormal density gives

$$f_{PT}(y) = \int_0^y \frac{1}{x\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log x - \mu)^2}{2\sigma^2}\right\} \frac{1}{(y-x)\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log(y-x) - \mu)^2}{2\sigma^2}\right\} dx.$$

This does not simplify to a standard closed-form lognormal density.

3.4 Path density for PQ

For the PQ path,

$$Y_{PQ} = X_1 + X_2 + X_3.$$

The density is a three-fold convolution:

$$f_{PQ}(y) = \int_0^y \int_0^{y-x_1} f_X(x_1;\mu,\sigma) f_X(x_2;\mu,\sigma) f_X(y-x_1-x_2;\mu,\sigma)\,dx_2\,dx_1, \qquad y > 0.$$

After substituting the lognormal density, this becomes

$$\begin{aligned} f_{PQ}(y) = \int_0^y \int_0^{y-x_1} &\frac{1}{x_1\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log x_1 - \mu)^2}{2\sigma^2}\right\} \\ &\times \frac{1}{x_2\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log x_2 - \mu)^2}{2\sigma^2}\right\} \\ &\times \frac{1}{(y-x_1-x_2)\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log(y-x_1-x_2) - \mu)^2}{2\sigma^2}\right\} dx_2\,dx_1. \end{aligned}$$

This is also not a convenient closed-form density.

3.5 Path density for CP

For the CP path,

$$Y_{CP} = |X_1 - X_2|.$$

Let

$$D = X_1 - X_2.$$

Because X_1 and X_2 are i.i.d., the distribution of D is symmetric around zero. For $y > 0$,

$$f_D(y) = \int_0^\infty f_X(x+y;\mu,\sigma) f_X(x;\mu,\sigma)\,dx.$$

Therefore,

$$f_{CP}(y) = f_{|D|}(y) = f_D(y) + f_D(-y) = 2\int_0^{\infty} f_X(x+y;\mu,\sigma) f_X(x;\mu,\sigma)\,dx, \qquad y > 0.$$

Substituting the lognormal density gives

$$\begin{aligned} f_{CP}(y) = 2\int_0^{\infty} &\frac{1}{(x+y)\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log(x+y)-\mu)^2}{2\sigma^2}\right\} \\ &\times \frac{1}{x\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\log x-\mu)^2}{2\sigma^2}\right\} dx, \qquad y > 0. \end{aligned}$$

Stage 4: why an exact lognormal extension is difficult

The exact lognormal extension preserves the epidemiologic logic of Vink's framework, but it creates a practical difficulty.

The PS component remains easy to evaluate. However, CP, PT, and PQ require numerical integration of lognormal difference or sum distributions. In an EM algorithm, these path probabilities must be evaluated repeatedly during the E-step and the M-step.

This creates several problems:

1. **High computational cost**: each likelihood evaluation becomes time-consuming and expensive, especially for PT and PQ.
2. **Limited return considering the study purpose**: the current project does not primarily aim to estimate the exact CP, PT, or PQ density. The primary aim is to recover the PS SI distribution while accounting for nuisance paths in the ICC data.

These reasons lead to the following methodological compromise.

Stage 5: a structurally constrained lognormal mixture model

5.1 Principles of the constrained lognormal mixture model

The exact lognormal Vink extension is conceptually attractive but computationally awkward. The scientific goal of the current simulation study is more focused: to examine whether the PS serial interval can be recovered accurately when ICC intervals arise from a mixture of latent transmission paths.

Therefore, instead of insisting that CP, PT, and PQ must be represented by their exact lognormal difference or convolution distributions, the four path-specific densities are approximated by ordered lognormal components.

This working model is not claiming that the true PT path is exactly lognormal, nor that the true CP path is exactly lognormal. Rather, it uses a set of lognormal components with related constraints to approximate the relative positions and shapes of the latent transmission paths:

- CP should tend to be located to the left of PS;
- PS is the primary target distribution;
- PT should tend to be located to the right of PS;

- PQ should tend to be located further right than PT;
- uncertainty should expand on the original time scale as the path moves further away from the index case.

Thus, the model is described as a **structurally constrained lognormal mixture model**, which can be shown as

$$f_Y(y) = \sum_{g \in \{CP, PS, PT, PQ\}} w_g f_g(y),$$

where f_g denotes the lognormal density used for path g in this working-model section.

5.2 Basic distributional form

The selected working distribution for each path is lognormal:

$$X_g \sim \text{Lognormal}(\mu_g, \sigma_g^2), \qquad g \in \{CP, PS, PT, PQ\}.$$

At this stage, the distributional family has been chosen, but the parameter constraints have not yet been fixed. The next question is how the four path-specific parameter pairs (μ_g, σ_g^2) should be related so that the model is both interpretable and consistent with the main simulation study plan.

Stage 6: choosing parameter constraints for the working model

After Stage 5 fixes the basic working distribution as lognormal, the next task is to choose how the path-specific lognormal parameters should be constrained.

This step should start from the study target. The final estimands of interest are still on the original time scale: the PS mean, variance, standard deviation, median, and selected quantiles. Therefore, a natural first attempt is to impose simple constraints directly on natural-scale moments. After deriving the log-scale parameters implied by those natural-scale constraints, we can then examine whether this is the most convenient parameterization for the working mixture model.

The purpose of this stage is to build a reasonable baseline working model that can later be evaluated in the simulation study.

6.1 General lognormal identities

If

$$X \sim \text{Lognormal}(\mu, \sigma^2),$$

then

$$E(X) = \exp\left(\mu + \frac{\sigma^2}{2}\right),$$

$$\text{Var}(X) = \left(e^{\sigma^2} - 1\right) e^{2\mu + \sigma^2},$$

and

$$\text{Median}(X) = e^{\mu}.$$

Conversely, if the natural-scale mean and variance are

$$m = E(X), \qquad v = \text{Var}(X),$$

then the corresponding log-scale parameters are

$$\sigma^2 = \log\left(1 + \frac{v}{m^2}\right),$$

and

$$\mu = \log(m) - \frac{1}{2}\sigma^2 = \log\left(\frac{m^2}{\sqrt{v + m^2}}\right).$$

Here, the term natural scale refers to the original time scale of the ICC interval or serial interval. Therefore, the natural-scale mean and variance are the mean and variance of the random variable X itself, not the mean and variance of $\log(X)$.

6.2 First attempt: imposing constraints directly on natural-scale moments

After choosing the lognormal family for the path-specific working model, the most direct idea is to impose constraints on the quantities that are ultimately of scientific interest. These quantities are defined on the original time scale, including the mean and variance of the PS serial interval.

Let each path-specific ICC component be approximated by a lognormal distribution,

$$X_g \sim \text{Lognormal}(\mu_g, \sigma_g^2), \qquad g \in \{CP, PS, PT, PQ\},$$

where μ_g and σ_g^2 are the mean and variance on the log scale.

A natural-scale parameterization starts by defining

$$m_g = E(X_g), \qquad v_g = \text{Var}(X_g).$$

Since the primary target is the PS serial interval, let

$$m_{PS} = m, \qquad v_{PS} = v.$$

For the baseline setting, impose the following natural-scale mean constraints:

$$m_{CP} = 0.5m, \qquad m_{PS} = m, \qquad m_{PT} = 2m, \qquad m_{PQ} = 3m.$$

As an initial attempt, suppose the four paths share the same natural-scale variance:

$$v_{CP} = v_{PS} = v_{PT} = v_{PQ} = v.$$

Then the implied log-scale parameters are obtained from the standard lognormal moment transformation.

For the CP component,

$$\sigma_{CP}^2 = \log\left(1 + \frac{v}{(0.5m)^2}\right) = \log\left(1 + \frac{4v}{m^2}\right),$$

and

$$\mu_{CP} = \log(0.5m) - \frac{1}{2}\sigma_{CP}^2.$$

For the PS component,

$$\sigma_{PS}^2 = \log\left(1 + \frac{v}{m^2}\right),$$

and

$$\mu_{PS} = \log(m) - \frac{1}{2}\sigma_{PS}^2.$$

For the PT component,

$$\sigma_{PT}^2 = \log\left(1 + \frac{v}{(2m)^2}\right) = \log\left(1 + \frac{v}{4m^2}\right),$$

and

$$\mu_{PT} = \log(2m) - \frac{1}{2}\sigma_{PT}^2.$$

For the PQ component,

$$\sigma_{PQ}^2 = \log\left(1 + \frac{v}{(3m)^2}\right) = \log\left(1 + \frac{v}{9m^2}\right),$$

and

$$\mu_{PQ} = \log(3m) - \frac{1}{2}\sigma_{PQ}^2.$$

Under this natural-scale constrained parameterization, the continuous mixture density is

$$f(x \mid \Theta) = w_{CP} f_{CP}(x) + w_{PS} f_{PS}(x) + w_{PT} f_{PT}(x) + w_{PQ} f_{PQ}(x),$$

where

$$w_{CP} + w_{PS} + w_{PT} + w_{PQ} = 1.$$

The lognormal density is

$$f_g(x) = \frac{1}{x\sigma_g\sqrt{2\pi}} \exp\left[-\frac{(\log x - \mu_g)^2}{2\sigma_g^2}\right], \qquad x > 0.$$

Therefore, after substituting the natural-scale moment constraints, the full baseline mixture density becomes

$$\begin{aligned}
f(x \mid \Theta) = w_{CP} &\frac{1}{x\sqrt{\log\left(1+\frac{4v}{m^2}\right)}\sqrt{2\pi}} \exp\left[-\frac{\left\{\log x - \left[\log(0.5m) - \frac{1}{2}\log\left(1+\frac{4v}{m^2}\right)\right]\right\}^2}{2\log\left(1+\frac{4v}{m^2}\right)}\right] \\
&+ w_{PS} \frac{1}{x\sqrt{\log\left(1+\frac{v}{m^2}\right)}\sqrt{2\pi}} \exp\left[-\frac{\left\{\log x - \left[\log(m) - \frac{1}{2}\log\left(1+\frac{v}{m^2}\right)\right]\right\}^2}{2\log\left(1+\frac{v}{m^2}\right)}\right] \\
&+ w_{PT} \frac{1}{x\sqrt{\log\left(1+\frac{v}{4m^2}\right)}\sqrt{2\pi}} \exp\left[-\frac{\left\{\log x - \left[\log(2m) - \frac{1}{2}\log\left(1+\frac{v}{4m^2}\right)\right]\right\}^2}{2\log\left(1+\frac{v}{4m^2}\right)}\right] \\
&+ w_{PQ} \frac{1}{x\sqrt{\log\left(1+\frac{v}{9m^2}\right)}\sqrt{2\pi}} \exp\left[-\frac{\left\{\log x - \left[\log(3m) - \frac{1}{2}\log\left(1+\frac{v}{9m^2}\right)\right]\right\}^2}{2\log\left(1+\frac{v}{9m^2}\right)}\right].
\end{aligned}$$

This formulation has the advantage of starting directly from the natural-scale estimands. However, the resulting mixture density is algebraically cumbersome. The constraints are simple on the natural scale, but the parameters that enter the lognormal density are nonlinear transformations of m and v.

In addition, the assumption of a common natural-scale variance across CP, PS, PT, and PQ maybe a strong modeling choice. If PT and PQ represent longer transmission paths, it may be more natural to allow their uncertainty on the original time scale to increase with path length.

For these reasons, a second parameterization way is considered, where the structural constraints are imposed directly on the log-scale parameters.

6.3 Second attempt: imposing constraints on log-scale parameters

The difficulty in Section 6.2 suggests that it may be more convenient to impose structural constraints directly on the log-scale parameters of the lognormal distribution.

This does not mean that our target of interest changes to the log scale. The final estimands remain the natural-scale summaries of the PS serial interval, including the PS mean, variance, standard deviation, median, and quantiles. The log scale is used mainly as a convenient modeling scale for specifying the mixture components.

For the baseline setting, let the four path-specific components be

$$X_{CP} \sim \text{Lognormal}(\mu + \log 0.5, \sigma^2),$$

$$X_{PS} \sim \text{Lognormal}(\mu, \sigma^2),$$

$$X_{PT} \sim \text{Lognormal}(\mu + \log 2, \sigma^2),$$

and

$$X_{PQ} \sim \text{Lognormal}(\mu + \log 3, \sigma^2).$$

Equivalently, this uses simple additive shifts on the log-scale location parameter:

$$\mu_{CP} = \mu + \log 0.5,$$

$$\mu_{PS} = \mu,$$

$$\mu_{PT} = \mu + \log 2,$$

and

$$\mu_{PQ} = \mu + \log 3.$$

The continuous mixture density is therefore

$$\begin{aligned} f(x \mid \Theta) = w_{CP} \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{\{\log x - (\mu + \log 0.5)\}^2}{2\sigma^2}\right] \\ + w_{PS} \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log x - \mu)^2}{2\sigma^2}\right] \\ + w_{PT} \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{\{\log x - (\mu + \log 2)\}^2}{2\sigma^2}\right] \\ + w_{PQ} \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{\{\log x - (\mu + \log 3)\}^2}{2\sigma^2}\right], \end{aligned}$$

with

$$w_{CP} + w_{PS} + w_{PT} + w_{PQ} = 1.$$

Compared with the natural-scale parameterization in Section 6.2, this model is much more compact. The fitted parameters are μ, σ, and the mixture weights, while the path-specific locations are generated through simple fixed shifts on the log scale.

The next question is what this log-scale constraint implies on the original time scale.

For a path-specific component with multiplier k, write

$$X_k \sim \text{Lognormal}(\mu + \log k, \sigma^2),$$

where, in the baseline setting,

$$k = 0.5 \text{ for CP}, \qquad k = 1 \text{ for PS}, \qquad k = 2 \text{ for PT}, \qquad k = 3 \text{ for PQ}.$$

The natural-scale mean is

$$E(X_k) = \exp\left(\mu + \log k + \frac{1}{2}\sigma^2\right) = k \exp\left(\mu + \frac{1}{2}\sigma^2\right).$$

Since

$$E(X_{PS}) = \exp\left(\mu + \frac{1}{2}\sigma^2\right),$$

we have

$$E(X_k) = kE(X_{PS}).$$

Therefore, although the constraints are imposed directly on the log-scale location parameters, they imply proportional relationships for the natural-scale means:

$$E(X_{CP}) = 0.5E(X_{PS}),$$

$$E(X_{PT}) = 2E(X_{PS}),$$

and

$$E(X_{PQ}) = 3E(X_{PS}).$$

The difference from Section 6.2 lies in the variance structure.

For the same component,

$$\mathrm{Var}(X_k) = \left(e^{\sigma^2} - 1\right) e^{2(\mu+\log k)+\sigma^2}.$$

Thus,

$$\mathrm{Var}(X_k) = k^2 \left(e^{\sigma^2} - 1\right) e^{2\mu+\sigma^2}.$$

Since

$$\mathrm{Var}(X_{PS}) = \left(e^{\sigma^2} - 1\right) e^{2\mu+\sigma^2},$$

we obtain

$$\mathrm{Var}(X_k) = k^2 \mathrm{Var}(X_{PS}).$$

Therefore, imposing common variance on the log scale implies that the natural-scale variance increases quadratically with the location multiplier:

$$\mathrm{Var}(X_{CP}) = 0.25\mathrm{Var}(X_{PS}),$$

$$\mathrm{Var}(X_{PT}) = 4\mathrm{Var}(X_{PS}),$$

and

$$\mathrm{Var}(X_{PQ}) = 9\mathrm{Var}(X_{PS}).$$

Equivalently, the natural-scale standard deviation scales in the same proportion as the natural-scale mean:

$$\mathrm{SD}(X_{CP}) = 0.5\mathrm{SD}(X_{PS}),$$

$$\mathrm{SD}(X_{PT}) = 2\mathrm{SD}(X_{PS}),$$

and

$$\mathrm{SD}(X_{PQ}) = 3\mathrm{SD}(X_{PS}).$$

This variance structure may be cleaner with the biological interpretation of longer transmission chains. If PT and PQ represent longer paths from the index case, the accumulated time interval is expected to be larger, and the uncertainty on the original time scale may also increase. The common log-scale variance assumption captures this by allowing variability to expand with path length.

The estimands remain on the natural scale. This parametraion gives a compact lognormal mixture model while still producing interpretable natural-scale mean and variance relationships.

For the PS component, the final natural-scale estimands are

$$E(X_{PS}) = \exp\left(\mu + \frac{1}{2}\sigma^2\right),$$

$$\mathrm{Var}(X_{PS}) = \left(e^{\sigma^2} - 1\right) e^{2\mu+\sigma^2},$$

$$\mathrm{SD}(X_{PS}) = \sqrt{\left(e^{\sigma^2} - 1\right) e^{2\mu+\sigma^2}},$$

and

$$\mathrm{Median}(X_{PS}) = \exp(\mu).$$

The p-th quantile of the PS distribution is

$$Q_p(X_{PS}) = \exp(\mu + \sigma z_p),$$

where z_p is the p-th quantile of the standard normal distribution.

Therefore, the baseline working model is specified through simple log-scale constraints, but its performance will be evaluated through natural-scale PS estimands. Whether this parameterization is reasonable enough will be tested through the simulation study.

Conclusion

The rationale leading to the current model setting can be summarized as follows:

1. Start from the original Vink normal/gamma path-mixture framework.
2. Recognize that PS is the primary estimand, while CP, PT, and PQ are nuisance path components that must be separated from PS.
3. Note that path separation depends on the distributional assumptions imposed on the mixture components.
4. Consider whether a right-skewed lognormal SI distribution may be more appropriate than a normal SI distribution.
5. Derive the exact lognormal extension of the Vink framework, including the integral forms for CP, PT, and PQ.
6. Recognize that the exact extension is mathematically valid but computationally inconvenient.
7. Introduce a structurally constrained lognormal mixture model as a baseline working model.
8. Start parameter selection from natural-scale moment constraints, then move to log-scale parameterization because it is simpler and reasonable biologically.
9. Report and evaluate the main estimands on the original time scale.
10. Use the simulation study to test whether this working model can recover the PS distribution accurately and robustly.